

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem Statement #05 - Campus & Academic Operations

Academic Elective Bidding & Allocation System

Architecture Selection: We chose Layered Architecture for the Academic Elective Bidding & Allocation System.

Architectural Choice: The system is organized into Presentation, Business and Data layers. The Presentation Layer contains the Student Portal and Admin Portal. The Business Layer contains the Authentication Service, Bidding & Allocation Service, Elective Management Service and Notification Service. The Data Layer contains the Academic Database.

Reason 1: Layered Architecture provides clear separation between the user interface, academic bidding and allocation logic, and data storage. This makes the system easier to maintain and modify.

Reason 2: Bidding and allocation rules can be centralized in the Business Layer, allowing student bids and allocation operations to be processed consistently.

Security Advantage: Authentication and authorization can be handled through the Authentication Service, while users can be prevented from directly accessing the academic database.

Performance Benefit: The system can optimize business operations and database queries independently, allowing frequent elective and bidding operations to be processed efficiently.

Architecture: Presentation Layer -> Business Layer -> Data Layer